

4. Two cards are selected at a random from a box which contains 5 cards numbered 1, 1, 2, 2, 3. Find Joint distribution of X and Y , where X denotes the sum and Y the maximum of the two numbers drawn. Also determine $\text{Cov}(X, Y)$ and $\rho(X, Y)$.

→ Possible selection: ${}^5C_2 = 10$

$C_1: C_2: C_3: C_4: C_5$
1 1 2 2 3

Let; X : Sum, Y : Maximum of two cards drawn

$S = \{(1, 1)(1, 2)(1, 2)(1, 3)(1, 2)(1, 2)(1, 3)(2, 2)(2, 3)(2, 3)\}$

$X = \{2, 3, 3, 4, 3, 3, 4, 4, 5, 5\}$

$Y = \{1, 2, 2, 3, 2, 2, 3, 2, 3, 3\}$

Range of X : $\{2, 3, 4, 5\}$, Range of Y : $\{1, 2, 3\}$

$X \backslash Y$	1	2	3	$f(x)$
2	$\frac{1}{10}$	0	0	$\frac{1}{10}$
3	0	$\frac{2}{5}$	0	$\frac{2}{5}$
4	0	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{3}{10}$
5	0	0	$\frac{1}{5}$	$\frac{1}{5}$
$g(y)$	$\frac{1}{10}$	$\frac{1}{2}$	$\frac{2}{5}$	$\sum_{ij} = 1$

Marginal Distribution of x :

$X:$	2	3	4	5
$f(x):$	$1/10$	$2/5$	$3/10$	$1/5$
$Y:$	1	2	3	
$g(y):$	$1/10$	$1/2$	$2/5$	

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

where; $E(X) = \sum x f(x) = 2(\frac{1}{10}) + 3(\frac{2}{5}) + 4(\frac{3}{10}) + 5(\frac{1}{5}) = 3.6$

$$E(Y) = \sum y g(y) = 1(\frac{1}{10}) + 2(\frac{2}{5}) + 3(\frac{1}{5}) = 2.3$$

$$E(XY) = \sum \sum x_i y_j f_{ij} = 2(1)(\frac{1}{10}) + 3(2)(\frac{2}{5}) + 4(2)(\frac{1}{10}) + 4(3)(\frac{1}{5}) + 5(3)(\frac{1}{5}) = 8$$

$$\therefore \text{Cov}(X, Y) = 8 - (3.6)(2.3) = \underline{\underline{0.28}}$$

5. A bag contains 3 white, 2 red, 2 green balls. Two balls are selected at random. If X & Y are RV denoting of white and red balls respectively. Determine:

- i) Joint Distribution of X and Y .
- ii) Marginal distribution of X and Y .
- iii) $\text{Cov}(X, Y)$

→ Let, X : no. of white balls.
 Y : no. of red balls.

{ 3 white + 2 red + 2 green
7 balls }

Possible selection: ${}^7C_2 = 21$

$$X = \{0, 1, 2, 3\}, Y = \{0, 1, 2\}$$

		10/35	20/35	5/35	
i)	$X \backslash Y$	0	1	2	
	0	0	2/35	2/35	4/35
	1	3/35	12/35	3/35	18/35
	2	6/35	6/35	0	12/35
	3	1/35	0	0	1/35

$$P(X=3, Y=0) = 3c_3 \times 2c_0 \times 2c_0 = 1/35$$

$$P(X=3, Y=1) = 0$$

$$P(X=3, Y=2) = 0$$

Marginal Distribution of X & Y:

$X:$	0	1	2	3	$Y:$	0	1	2
$f(x):$	4/35	18/35	12/35	1/35	$g(y):$	10/35	20/35	5/35

$$\text{iii) } \text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$\text{where; } E(X) = \sum x f(x) = 18/35 + 24/35 + 3/35 = 45/35 = 9/7$$

$$E(Y) = \sum y g(y) = 20/35 + 10/35 = 30/35 = 6/7$$

$$E(XY) = \sum \sum x_i y_j T_{ij} = 1(1)(12/35) + 2(3/35) + 2(6/35) + 0$$

$$\therefore \text{Cov}(X, Y) = 6/7 - 54/49 = -0.24$$

Joint Probability Distribution for Continuous Random Variables:

If 'X' and 'Y' are the continuous random variables then $F(X, Y) = P(X=x, Y=y)$ is called Joint probability density function of the random variables 'X' and 'Y' where

$$F(X, Y) \geq 0 \text{ and } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(x, y) dx dy = 1$$

$$\text{Note: } P(X=a, Y=b) \Rightarrow P(a \leq x \leq b, c \leq y \leq d) = \int_a^b \int_c^d F(x, y) dy dx$$

Expectation, Mean, Covariance, Correlation:

$$\text{Mean; } \mu_x = E(X) = \int_{-\infty}^{\infty} x f(x) dx, \mu_y = E(Y) = \int_{-\infty}^{\infty} y g(y) dy$$

$$\mu_{xy} = E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy F(x, y) dx dy$$

$$\text{Variance; } \sigma_x^2 = V(X) = E(X^2) - [E(X)]^2$$

$$\sigma_y^2 = V(Y) = E(Y^2) - [E(Y)]^2$$

4.7 Covariance, $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$

Correlation, $\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$

8. Marginal distribution functions of 'x' and 'y'

$$f(x) = \int_y F(x, y) dy \quad ; \quad g(y) = \int_x F(x, y) dx$$

Let

1. Suppose that the joint density function of two continuous random variables X and Y is

$$f(x, y) = \begin{cases} x^2 + xy/3, & 0 < x < 1, 0 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

Find:

i) $P(X > 1/2)$

ii) $P(0 < x < 1/2)$

iii) $E(X)$, $E(Y)$ and $E(XY)$

iv) Marginal distribution of X and Y

$$\rightarrow \text{i) } P(X > 1/2) = \int_{x=1/2}^1 \int_{y=0}^2 (x^2 + \frac{xy}{3}) dy dx \Rightarrow \int_{x=1/2}^1 (x^2 y + \frac{xy^2}{6}) \Big|_0^2 dx$$

$$\Rightarrow \int_{x=1/2}^1 (x^2(2) + \frac{2(x)^2}{6} - 0) dx \Rightarrow \left[\frac{2x^3}{3} + \frac{x^3}{3} \right]_{1/2}^1$$

$$\text{Cor } P(X > 1/2) = \left[\frac{2}{3} + \frac{1}{6} - \left(\frac{1}{12} + \frac{1}{48} \right) \right] \Rightarrow \left[\frac{2}{3} \left(x^3 + \frac{x^3}{3} \right) \right]_{1/2}^1$$

$$P(X > 1/2) = \left[\frac{2}{3} \left(\left(1 - \frac{1}{8} \right) \right) + \left(\frac{1}{3} \left[1 - \frac{1}{4} \right] \right) \right]$$

$$= \left(\frac{2}{3} \left(\frac{7}{8} \right) + \frac{0.75}{3} \right) = \underline{\underline{0.833}}$$

$$\text{ii) } P(0 < x < 1/2, 0 < y < 1) = \int_{x=0}^{1/2} \int_{y=0}^1 (x^2 + \frac{xy}{3}) dy dx$$

$$\Rightarrow \int_{x=0}^{1/2} \left(\frac{x^2 y + \frac{xy^2}{6} \right) \Big|_0^1 dx \Rightarrow \int_{x=0}^{1/2} \left(x^2 + \frac{x}{6} \right) dx \Rightarrow \left[\frac{x^3}{3} + \frac{x^2}{12} \right]_{0}^{1/2}$$

$$P(0 < x < 1/2, 0 < y < 1) = \left[\frac{1}{24} + \frac{1}{48} \right] = \underline{\underline{0.416}}$$

ii) iii) Marginal distribution of X

$$\rightarrow f(x) = \int_y F(x, y) dy \Rightarrow \int_{y=0}^2 (x^2 + \frac{xy}{3}) dy \Rightarrow \left[x^2 y + \frac{xy^2}{6} \right]_{0}^2$$

$$f(x) = 2x^2 + \frac{2x}{3}$$

1) Marginal Distribution of Y:

$$g(y) = \int_0^1 (x^2 + \frac{xy}{3}) dx \Rightarrow g(y) = \left[\frac{x^3}{3} + \frac{x^2 y}{6} \right]_0^1 \Rightarrow \frac{1}{3} + \frac{y}{6}$$

$$E(X) = \int_0^1 x f(x) dx \Rightarrow \int_0^1 x (2x^2 + \frac{2x}{3}) dx \Rightarrow \int_0^1 (2x^3 + \frac{2x^2}{3}) dx$$

$$\Rightarrow \left[\frac{2x^4}{4} + \frac{2x^3}{9} \right]_0^1 \Rightarrow \frac{1}{2} + \frac{2}{9} \Rightarrow E(X) = 0.722$$

$$E(Y) = \int_0^2 y g(y) dy \Rightarrow \int_0^2 y (\frac{1}{3} + \frac{y}{6}) dy \Rightarrow \int_0^2 (\frac{y}{3} + \frac{y^2}{6}) dy$$

$$\Rightarrow \left[\frac{y^2}{6} + \frac{y^3}{18} \right]_0^2 \Rightarrow \frac{4}{6} + \frac{8}{18} \Rightarrow E(Y) = 1.11$$

$$E(XY) = \int_0^1 \int_0^2 xy (x^2 + \frac{xy}{3}) dy dx \Rightarrow \int_0^1 \left(\frac{x^3 y^2}{2} + \frac{x^2 y^3}{9} \right) dx$$

$$\Rightarrow \int_0^1 \left(\frac{4x^3}{3} + \frac{8x^2}{9} \right) dx \Rightarrow \left[\frac{4x^4}{4} + \frac{8x^3}{27} \right]_0^1 \Rightarrow 0.5 + 0.296$$

$$E(XY) = 0.796$$

$$\therefore \text{Cov}(X, Y) = E(XY) - E(X)E(Y) \Rightarrow 0.796 - (0.722)(1.11) \Rightarrow -0.0006$$

2) Find the value of 'k' from the joint probability density function of X and Y given by,

$$f(x, y) = \begin{cases} (6-x-y)k, & 0 < x < 2, 0 < y < 4 \\ 0, & \text{otherwise} \end{cases}$$

→ Given that, $f(x, y)$ is a pdf

$$\therefore f(x, y) \geq 0 \text{ and } \int_0^2 \int_0^4 f(x, y) dx dy = 1$$

$$\Rightarrow \int_0^2 \int_0^4 (6-x-y)k dy dx = 1 \Rightarrow k \int_0^2 \left(6y - xy - \frac{y^2}{2} \right) dx$$

$$\Rightarrow k \int_0^2 (24 - 4x - 8) dx = 1 \Rightarrow k \left[24x - \frac{4x^2}{2} - 8x \right]_0^2 = 1$$

$$\Rightarrow k [48 - 16 - 16] = 1 \Rightarrow k = 0.0416$$

Find $P(X < 1, Y < 3)$:

$$k \int_0^1 \int_0^3 (6y - xy - \frac{y^2}{2}) dx \Rightarrow k \int_0^3 18 - 3x - \frac{y^2}{2} dx$$

$$\Rightarrow k \left[18x - \frac{3x^2}{2} - 9x/2 \right]_0^2 \Rightarrow k [18 - 9 - 4.5] \Rightarrow k [4.5] \Rightarrow 0.4992$$

Find $P(x+y < 3)$

$$x \rightarrow 0, 2$$

$$y \rightarrow 0, 3-x$$

$$P(x+y < 3) = \int_0^2 \int_0^{3-x} k(6-x-y) dy dx$$

$$\Rightarrow k \int_0^2 \left(6y - xy - \frac{y^2}{2} \right)_{y=0}^{y=3-x} dx$$

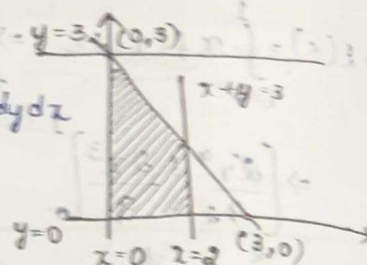
$$\Rightarrow k \int_0^2 \left(6(3-x) - (3-x)x - \frac{(3-x)^2}{2} \right) dx \Rightarrow k \int_0^2 \left(18 - 6x - 3x + x^2 - \frac{9 - 6x + x^2}{2} \right) dx$$

$$\Rightarrow k \int_0^2 \left(18 - 6x - 3x + xy - \left[\frac{9}{2} + x^2 - 6x \right] \right) dx \Rightarrow x^2 - 12x - 3y + 6 + xy$$

$$\Rightarrow (36 - 12x - 6y + 2xy - 9 - x^2 + 6x) dx$$

$$\Rightarrow \frac{k}{2} \int_0^2 (-x^2 - 6y + 2xy - 6x + 27) dx \Rightarrow \frac{k}{2} \left[-\frac{x^3}{3} - 6xy + \frac{2x^2y}{2} - \frac{6x^2}{2} + 27x \right]_0^2$$

$$\Rightarrow \frac{k}{2} \left[-\frac{8}{3} - 12y + 4y - 12 + 54 \right] \Rightarrow P(x+y < 3) = 0.5$$



Markov Chain and Queuing TheoryMarkov Process:Stochastic Process (Random Process):

A stochastic process is a collection of random variables $\{x(s, t)\}$ where $s \in S$, $t \in T$. that are the functions of real variable namely 'Type T'. The values assumed by the random variable $x(t)$ are called states. The set of all possible values forms state space of the process.

If the state space of the stochastic process is discrete then it is called discrete state process also called a chain.

Markov Process:

also called memory-less process.

If the future behaviour of the process depends only on the present state but not on the past then it is called Markov Process. The discrete parameters of Markov process is called Markov Chain.

Markov Chain:

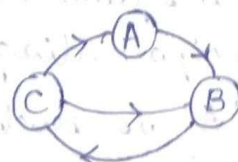
A sequence of repeated trials of an experiment with the outcome at any step in the sequence depends almost on the outcome of the preceding step and not on any previous outcomes.

Example: Three boys A, B & C throwing a ball to each other. A always throws the ball to B, B always throws to C. C is however just as likely to throw ball to B. Here the boy throwing the ball is not influenced by those who previously had the ball.

→ Here A, B, C are called states.

∴ State Space = $\{A, B, C\}$

Transition
Digraph



Transition Matrix:

$$P = \begin{matrix} & \begin{matrix} \text{A} & \text{B} & \text{C} \end{matrix} \\ \begin{matrix} \text{A} \\ \text{B} \\ \text{C} \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} \end{matrix} \rightarrow \text{immediate next state}$$

↓
Present State

• Probability Vector: A vector $V = \{v_1, v_2, \dots, v_n\}$ is called probability vector if and only if it satisfies the following conditions;

i) $v_i \geq 0 \quad \forall i$ & ii) $\sum_{i=1}^n v_i = 1$

Example: $\{1, 0\}$, $\{1/2, 1/2, 0\}$, $\{1/3, 1/3, 1/3\}$, etc.

• Stochastic Matrix: (Probabilistic Matrix)

A Square Matrix $P = [P_{ij}]$ is called a probabilistic matrix if each row is a probability vector, i.e., each entry of 'P' is non negative and sum of the entries in each row is equal to '1'.

Example: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 \\ 1/2 & 1/2 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$ etc.

• Regular stochastic matrix: A stochastic matrix 'P' is said to be regular stochastic if & only if all the entries of some power of P^n are all positive. ($P^n > 0$).

Example: $P = \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix} \Rightarrow P^2 = \begin{bmatrix} 1/2 & 1/2 \\ 1/4 & 1/4 \end{bmatrix}$

Hence, 'P' is regular stochastic matrix (RSM).

ii) $P = \begin{bmatrix} 1 & 0 \\ 1/2 & 1/2 \end{bmatrix}$ is not a RSM.

Here, some entries in the matrices P^2, P^3, P^4 will contain zeros.

NOTE:

Properties of Regular Stochastic Matrix:

→ If a square matrix is said to be RSM, if the principal diagonal does not contain '1'.

→ If 'A' is stochastic matrix then there exists a probability vector 'V' such that;

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i) $V_i > 0 \forall i$

ii) $VP = V$

1. Find the unified probability vector for the following matrices:

i) $P = \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix}$

called the

→ we need to find $V = [V_1, V_2]$ such that $\sum V = 1$ { $V_1 + V_2$ } & $VP = V$

Consider; $VP = V$

$$\Rightarrow [V_1 \ V_2] \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix} = [V_1 \ V_2]$$

$$\Rightarrow [V_2/2 \quad V_1 + V_2/2] = [V_1 \ V_2] \Rightarrow \frac{V_2}{2} = V_1 \text{ and } V_1 + \frac{V_2}{2} = V_2$$

we have; $V_1 + V_2 = 1$

$$\Rightarrow V_2 = 2V_1 \Rightarrow V_1 + \frac{2V_1}{2} = V_2$$

$$V_1 + 2V_1 = 1$$

$$V_1 = 1/3$$

$$V_2 = 2(1/3) = 2/3$$

$\therefore V = [1/3 \ 2/3]$ is the unified probability vector.

[It is a RSM]

ii) $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix}$

→ we need to find $V = [V_1 \ V_2 \ V_3]$ such that $\sum V = 1$ { $V_1 + V_2 + V_3$ } and $VP = V$.

Consider; $VP = V$

$$\Rightarrow [V_1 \ V_2 \ V_3] \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} = [V_1 \ V_2 \ V_3]$$

$$\Rightarrow [V_3/2 \quad V_1 + V_3/2 \quad V_2] = [V_1 \ V_2 \ V_3]$$

$$\Rightarrow V_1 = \frac{V_3}{2}, \quad V_2 = V_1 + \frac{V_3}{2}, \quad V_3 = V_2 \Rightarrow V_3 = 2V_1$$

$$\text{we've ; } V_1 + V_2 + V_3 = 1 \Rightarrow V_1 = \frac{2V_2 - V_3}{2}$$

$$\left[\frac{V_3}{2} + V_1 + \frac{V_3}{2} + V_2 = 1 \right]$$

$$\therefore V_3 = 2V_1 \Rightarrow \frac{2V_1}{2} + V_1 + \frac{2V_1}{2} + 2V_1 = 1 \therefore V_1 + V_2 + V_3 = 1$$

$$V_1 = 1/5$$

$$V_2 = 1 - \frac{1}{5} - \frac{2}{5}$$

$$V_2 = \frac{2}{5}$$

$$\therefore V_3 = 2V_1 \Rightarrow V_3 = \frac{2}{5}$$

2. Verify whether the following matrices are regular or not.

1.
$$P = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$\rightarrow P^2 = \begin{bmatrix} 0.54 & 0.06 & 0.4 \\ 1 & 0 & 0 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}$

$P^3 = \begin{bmatrix} 0.508 & 0.162 & 0.33 \\ 0.2 & 0.3 & 0.5 \\ 0.54 & 0.06 & 0.4 \end{bmatrix}$

Here all entries of P^2 and $P^3 > 0$. Hence, P is a regular stochastic matrix.

2.
$$P = \begin{bmatrix} 1 & 0 & 0 \\ 1/3 & 1/3 & 1/3 \\ 2/3 & 1/3 & 0 \end{bmatrix}$$

Here we've 1 in the diagonal in $[P]$. Hence, it is not a regular stochastic matrix.

Transition Probability and Transition Matrix:

The probability ' P_{ij} ' is associated with every pair of states (a_i, a_j) that ' a_j ' occurs immediately after ' a_i ' occurs. Such a stochastic process is called Finite Markov Chain. If each $P_{ij} > 0$ then it is called Transition Probability and the matrix formed from Transition Probability is called Transition Matrix.

$$P = P_{ij} = \begin{bmatrix} P_{11} & P_{12} & \dots & P_{1m} \\ P_{21} & P_{22} & \dots & P_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ P_{m1} & P_{m2} & \dots & P_{mm} \end{bmatrix}$$

If the system is in state ' a_i ' then the row vector $a_i [P_{i1} \ P_{i2} \ \dots \ P_{im}]$ represents the probability of all possible outcomes at the next trial.

Higher Transition Probabilities:

The probability that a system changes from the state a_i to the state a_j in exactly ' n ' steps is denoted by $P_{ij}^{(n)}$ or $p^{(n)}$. The matrix formed by the probabilities $P_{ij}^{(n)}$ is called ' n step transition matrix.'

The n^{th} step transition matrix = n^{th} power of P .
 $P^{(n)} = P^n$

Note:

Let P be the TPM of the Markov chain and $P_i = (P_{i1}, P_{i2}, \dots, P_{im})$ be the probability distribution at some arbitrary time. Then $(P_i P, P_i P^2, \dots, P_i P^n)$ respectively are the probabilities of the system after one step, two steps, ..., n steps.

Let, $P^{(0)} = P_1^{(0)}, P_2^{(0)}, \dots, P_n^{(0)}$ denotes the initial probability distribution and $P^{(n)} = P_1^{(n)}, P_2^{(n)}, \dots, P_n^{(n)}$ denotes the n^{th} step probability distribution then, $P_1^{(1)} = P^{(0)} P$, $P_1^{(2)} = P^{(1)} P / P_1^{(0)} P^2$, $P_1^{(3)} = P^{(0)} P^3$... so on. Thus, $P^{(n)} = P^{(0)} P^n$ or $P^{(n)} = P^{(n-1)} P$

Markov chain is said to be irreducible if the associated tpm is regular. (tpm $\rightarrow$ transition probability matrix).

1. Three boys A, B, C are throwing a ball to each other. A always throws the ball to B and B always to C. C is just as likely to throw ball to B as to A. If C was the first person to throw the ball then find the probability that after 4 throws i) A has the ball, ii) B has the ball and iii) C has the ball.

$\rightarrow$ State space = $\{A, B, C\}$

$$\text{tpm } P = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} \end{matrix}$$

Initial probability $\Rightarrow P^{(0)} = [0 \ 0 \ 1]$ $\{\because C \text{ throws the ball first}\}$
 After $P^{(4)}$ throws; $P^{(4)} = P^{(0)} P^4$

$$\Rightarrow P^{(4)} = [0 \ 0 \ 1] \begin{bmatrix} 1 & 0.5 & 0.5 \\ 0.25 & 0.25 & 0.5 \\ 0.25 & 0.5 & 0.25 \end{bmatrix} = [1/4 \ 1/2 \ 1/4]$$

$$\Rightarrow P^{(4)} = \begin{bmatrix} 0.25 & 0.25 & 0.5 \\ 0.25 & 0.5 & 0.25 \\ 0.625 & 0.375 & 0.5 \end{bmatrix}$$

After 4 throws; Probability (A) has ball = $\underline{1/4}$

(B) has ball = $\underline{1/2}$

(C) has ball = $\underline{1/4}$

2. A student's study habits are as follows. If he studies one night, he's 70% sure not to study the next night. On the other hand if he doesn't study one night he's 60% sure not to study the next night as well. How often does he study?

→ State Space: $\{S, N\}$
 $S \rightarrow \text{Study}$
 $N \rightarrow \text{Not Study}$

$$P = \begin{matrix} & \begin{matrix} S & N \end{matrix} \\ \begin{matrix} S \\ N \end{matrix} & \begin{bmatrix} 0.3 & 0.7 \\ 0.4 & 0.6 \end{bmatrix} \end{matrix}$$

To find the happening in the long run, we should find a fixed probability vector 'V' such that $\sum V = V_1 + V_2 = 1$ and $VP = V$

$$\Rightarrow [V_1 \ V_2] \begin{bmatrix} 0.3 & 0.7 \\ 0.4 & 0.6 \end{bmatrix} = [V_1 \ V_2]$$

$$\Rightarrow [0.3V_1 + 0.4V_2 \quad 0.7V_1 + 0.6V_2] = [V_1 \ V_2]$$

$$\Rightarrow \begin{cases} 0.3V_1 + 0.4V_2 = V_1 \\ 0.7V_1 + 0.6V_2 = V_2 \end{cases} \quad \begin{cases} V_1 + V_2 = 1 \end{cases}$$

$$\Rightarrow 0.3V_1 - V_1 = -0.4V_2$$

$$\Rightarrow -0.7V_1 = -0.4V_2$$

$$\Rightarrow V_1 = \frac{0.4}{0.7} V_2$$

$$\Rightarrow \frac{0.4}{0.7} V_2 + V_2 = 1$$

$$\Rightarrow 0.4V_2 + 0.7V_2 = 0.7$$

$$\Rightarrow 0.11V_2 = 0.7 \Rightarrow V_2 = \frac{7}{11}$$

$$\therefore V = \begin{bmatrix} 4/11 & 7/11 \end{bmatrix}$$

The probability that the student studies is $4/11$

3. Two boys b_1 and b_2 and two girls g_1 and g_2 are throwing a ball from one to the other. Each boy throws the ball to the other with probability $1/2$ and to each girl with probability $1/4$. On the other hand, each girl throws the ball to each boy with probability $1/2$ and never to the girl. In the long run, how often does each receive the ball?

→ State Space: $\{b_1, b_2, g_1, g_2\}$

$$t_{pm} = p =$$

$$\begin{matrix} & b_1 & b_2 & g_1 & g_2 \\ \begin{matrix} b_1 \\ b_2 \\ g_1 \\ g_2 \end{matrix} & \begin{bmatrix} 0 & 1/2 & 1/4 & 1/4 \\ 1/2 & 0 & 1/4 & 1/4 \\ 1/2 & 1/2 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \end{bmatrix} \end{matrix}$$

To find the happening in the long run, we should find the unified probability vector 'V' such that $\Sigma V = 1$ and $VP = V$

$$\Rightarrow [V_1 \ V_2 \ V_3 \ V_4] \begin{bmatrix} 0 & 1/2 & 1/4 & 1/4 \\ 1/2 & 0 & 1/4 & 1/4 \\ 1/2 & 1/2 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \end{bmatrix} = [V_1 \ V_2 \ V_3 \ V_4]$$

$$\Rightarrow \left[\frac{V_2 + V_3 + V_4}{2} \quad \frac{V_1 + V_3 + V_4}{2} \quad \frac{V_1 + V_2}{4} \quad \frac{V_1 + V_2}{4} \right] = [V_1 \ V_2 \ V_3 \ V_4]$$

$$\Rightarrow V_1 = \frac{V_2 + V_3 + V_4}{2}, \quad V_2 = \frac{V_1 + V_3 + V_4}{2}, \quad V_3 = \frac{V_1 + V_2}{4}, \quad V_4 = \frac{V_1 + V_2}{4} \Rightarrow V_1 + V_2 = 4V_3$$

$$\Rightarrow V_1 + V_2 + V_3 + V_4 = 1$$

$$\Rightarrow 4V_3 + V_3 + \frac{V_1 + V_2}{4} = 1$$

$$\Rightarrow 5V_3 + \frac{4V_3}{4} = 1 \Rightarrow V_3 = \frac{1}{5}$$

$$\Rightarrow V_1 + V_2 = \frac{4}{5}$$

$$\therefore V_1 + V_2 + V_3 + V_4 = 1$$

$$\Rightarrow \frac{4}{5} + \frac{1}{5} + V_4 = 1$$

$$\Rightarrow V_4 = 1 - \left(\frac{4}{5} + \frac{1}{5} \right)$$

$$\Rightarrow V_4 = 1 - 1 = 0$$

$$\text{But } \frac{V_1 + V_2}{4} = V_3$$

$$\frac{4}{5} = V_3 = \frac{1}{5}$$

$$V_4 = \frac{1}{5}$$

$$\therefore V_1 = \frac{V_2 + 1/5 + 1/5}{2}, \quad V_2 = \frac{V_1 + 1/5 + 1/5}{2}$$

$$V_1 = \frac{V_1 + 1/5 + 1/5 + 1/5 + 1/5}{2}$$

$$\Rightarrow 2V_1 - V_1 = 4/5 \Rightarrow V_1 = 4/5$$

$$\text{Thus; } V_1 + V_2 = \frac{4}{5}$$

$$\Rightarrow V_2 = 0$$

$$\therefore [V_1 \ V_2 \ V_3 \ V_4] = [4/5 \ 0 \ 1/5 \ 1/5]$$

4) 12. A salesman's territory consists of three cities A, B & C. He never sells in the same city on successive days. If he sells in city A, then the next day he sells in city B. If he sells in either B or C then the next day he is thrice as likely to sell in city A as in other city. In the long run how often does he sell in each of the cities?

→ State Space: $\{A, B, C\}$

$$1pm = P = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 3/4 & 0 & 1/4 \\ 3/4 & 1/4 & 0 \end{bmatrix} \end{matrix}$$

To find the happening in the long run, we should find UFPV 'V' such that $\sum V = V_1 + V_2 + V_3 = 1$ & $VP = V$

$$\Rightarrow [V_1 \ V_2 \ V_3] \begin{bmatrix} 0 & 1 & 0 \\ 3/4 & 0 & 1/4 \\ 3/4 & 1/4 & 0 \end{bmatrix} = [V_1 \ V_2 \ V_3]$$

$$\Rightarrow \left[\frac{3V_2 + 3V_3}{4} \quad \frac{V_1 + V_3}{4} \quad \frac{V_2}{4} \right] = [V_1 \ V_2 \ V_3]$$

$$\Rightarrow \frac{3V_2 + 3V_3}{4} = V_1, \quad \frac{V_1 + V_3}{4} = V_2, \quad \frac{V_2}{4} = V_3 \Rightarrow V_2 = 4V_3$$

$$\therefore V_1 + V_2 + V_3 = 1$$

$$\frac{3[4V_3] + 3V_3}{4} + 4V_3 + V_3 = 1$$

$$\Rightarrow \frac{15V_3 + 3V_3}{4} = 1 \Rightarrow 35V_3 = 4$$

$$\Rightarrow V_3 = 4/35$$

$$\Rightarrow V_2 = \frac{4 \times 4}{35} = \frac{16}{35}$$

$$\therefore V_1 = 1 - V_2 - V_3$$

$$\Rightarrow V_1 = 1 - \frac{16}{35} - \frac{4}{35}$$

$$\Rightarrow V_1 = \frac{35 - 16 - 4}{35} = \frac{15}{35} = \frac{3}{7}$$

We get; $V_1 = 0.43$, $V_2 = 0.46$, $V_3 = 0.11$

$$V = \begin{bmatrix} 0.43 & 0.46 & 0.11 \\ A & B & C \end{bmatrix}$$

In the long run, the probability that the salesman sells in city A = 0.43

city B = 0.46

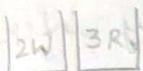
city C = 0.11

5) There are two white marbles in bag A and 3 red marbles in bag B. At each step of the process a marble is selected at random from each bag and the two marbles selected are interchanged, then find:

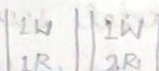
i) Transition probability matrix.

ii) What is the probability that there are 2 red marbles in 'A' after ^{three} two steps.

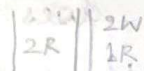
iii) In the long run what is the probability that there are two white marbles in bag 'A'.



S1



S2



S3

→ State Space: $\{S_1, S_2, S_3\}$

$$i) \quad \text{tpm} = P = \begin{matrix} & \begin{matrix} S_1 & S_2 & S_3 \end{matrix} \\ \begin{matrix} S_1 \\ S_2 \\ S_3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{bmatrix} \end{matrix}$$

$$S_2 \rightarrow S_1 \Rightarrow \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

$$S_2 \rightarrow S_2 \Rightarrow \frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times \frac{1}{3} = \frac{1}{2}$$

$$S_2 \rightarrow S_3 \Rightarrow \frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$$

$$S_3 \rightarrow S_2 \Rightarrow \frac{2}{2} \times \frac{2}{3} = \frac{2}{3}$$

$$S_3 \rightarrow S_3 \Rightarrow \frac{2}{2} \times \frac{1}{3} = \frac{1}{3}$$

$$ii) \quad p^{(3)} = p^{(0)} p^3$$

$$\text{Here; } p^{(0)} = [1 \ 0 \ 0]$$

$$p^{(3)} = [1 \ 0 \ 0] \begin{bmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{bmatrix}^3 \Rightarrow [1 \ 0 \ 0] \begin{bmatrix} 0.0833 & 0.6388 & 0.2777 \\ 0.1064 & 0.5879 & 0.3055 \\ 0.0925 & 0.6111 & 0.2962 \end{bmatrix}$$

$$p^{(3)} = \begin{bmatrix} 0.0833 & 0.6388 & 0.2777 \end{bmatrix} \text{ or } \begin{bmatrix} 1/12 & 23/36 & 5/18 \end{bmatrix}$$

The probability that there are 2 red marbles in A = 0.0833

iii) To find the happening in the long run, we should find

UFVP of 'V' such that $\sum V = V_1 + V_2 + V_3 = 1$ & $VP = V$

$$\Rightarrow [V_1 \ V_2 \ V_3] \begin{bmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{bmatrix} = [V_1 \ V_2 \ V_3]$$

$$\Rightarrow [V_2/6 \quad V_1 + V_2/6 + 2V_3/3 \quad V_2/3 + V_3/3] = [V_1 \ V_2 \ V_3]$$

$$\Rightarrow V_1 = \frac{V_2}{6} \Rightarrow 6V_1 = V_2, \quad V_2 = V_1 + \frac{V_2}{6} + \frac{2V_3}{3}, \quad V_3 = \frac{V_3 + V_2}{3} \Rightarrow \frac{2}{3}V_3 = \frac{V_2}{3}$$

$$\Rightarrow V_1 = \frac{V_2}{6} - \frac{V_2}{6} - \frac{2V_3}{3} \Rightarrow V_1 = \frac{5V_2}{6} - \frac{2V_3}{3} \Rightarrow V_2 = 2V_3$$

$$\Rightarrow V_1 = \frac{5 \cdot 2V_3}{6} - \frac{2V_3}{3} = \frac{5V_3}{3} - \frac{2V_3}{3} = \frac{3V_3}{3} = V_3$$

$$\therefore V_1 + V_2 + V_3 = 1 \Rightarrow V_3 + 2V_3 + V_3 = 1 \Rightarrow 4V_3 = 1$$

$$\Rightarrow V_3 = \frac{1}{4}$$

$$V_2 = 2 \left(\frac{1}{4} \right) = \frac{1}{2}$$

$$V_1 \Rightarrow V_2/6 = \frac{1/2}{6} = \frac{1}{12}$$

$$V = \begin{bmatrix} 1/12 & 1/2 & 1/4 \\ S_1 & S_2 & S_3 \end{bmatrix}$$

$\therefore$ Probability that there are 2 white marbles in 'A' = 1/12

6. In machine learning, a model's inference habits on a binary classification task follow a Markov chain pattern, if it performs well on one batch, it is 67% likely not to perform well on the next batch, if it performs poorly, it is 58% likely to perform poorly on the next batch as well. In the long run, what

proportion of batches does the model successfully infer

→ State Space: $\{W, N\}$

$W \rightarrow$ well, $N \rightarrow$ not well.

$$tpm = P = \begin{matrix} & \begin{matrix} W & N \end{matrix} \\ \begin{matrix} W \\ N \end{matrix} & \begin{bmatrix} 0.33 & 0.67 \\ 0.42 & 0.58 \end{bmatrix} \end{matrix}$$

To find the happening in the long run, we should find UFPV 'V' such that $\sum V = V_1 + V_2 = 1$ and $VP = V$

$$\Rightarrow [V_1, V_2] \begin{bmatrix} 0.33 & 0.67 \\ 0.42 & 0.58 \end{bmatrix} = [V_1, V_2]$$

$$\Rightarrow [0.33V_1 + 0.42V_2, 0.67V_1 + 0.58V_2] = [V_1, V_2]$$

$$\Rightarrow V_1 = 0.33V_1 + 0.42V_2, \quad 0.67V_1 + 0.58V_2 = V_2$$

$$\Rightarrow V_1 - 0.33V_1 = 0.42V_2 \quad \Rightarrow V_2 - 0.58V_2 = 0.67V_1$$

$$= 0.67V_1 = 0.42V_2 \quad \quad \quad 0.42V_2 = 0.67V_1$$

$$V_1 = \frac{42V_2}{67} \quad \quad \quad V_2 = \frac{67V_1}{42}$$

$$\therefore \frac{42V_2}{67} + V_2 = 1 \Rightarrow \frac{109V_2}{67} = 1 \Rightarrow V_2 = \frac{67}{109}$$

$$\therefore V_1 = \frac{42}{67} \times \frac{67}{109} = \frac{42}{109}$$

$$\text{we get; } V = \left[\frac{42}{109}, \frac{67}{109} \right]$$

The proportion of batches does the model successfully are; 0.3853

7. A player has Rs. 300. At each play of a game, he loses Rs. 100 with probability $3/4$ but wins Rs. 200 with probability $1/4$. He stops playing if he has lost his Rs. 300 or he has won atleast Rs. 300.

i) Determine tpm of Markov Chain.

ii) Find the probability that there are atleast 4 plays to the game.

→ 100 → $3/4$ (loses)

200 → $1/4$ (wins)

State Space $\{S.S\} = \{0, 100, 200, 300, 400, 500, 600\}$

$$\begin{matrix} & \begin{matrix} 0 & 100 & 200 & 300 & 400 & 500 & 600 \end{matrix} \\ \begin{matrix} 0 \\ 100 \\ 200 \\ 300 \\ 400 \\ 500 \\ 600 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3/4 & 0 & 0 & 1/4 & 0 & 0 & 0 \\ 0 & 3/4 & 0 & 0 & 1/4 & 0 & 0 \\ 0 & 0 & 3/4 & 0 & 0 & 1/4 & 0 \\ 0 & 0 & 0 & 3/4 & 0 & 0 & 1/4 \\ 0 & 0 & 0 & 0 & 3/4 & 0 & 1/4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix} = tpm = P$$

$$ii) p^{(4)} = p^{(3)} p^4 ; p^{(1)} = p^{(0)} p ; p^{(2)} = p^{(1)} p ; p^{(3)} = p^{(2)} p ; p^{(4)} = p^{(3)} p$$

$$\therefore p^{(0)} = [0, 0, 0, 1, 0, 0, 0]$$

$$p^{(1)} = [0, 0, 0, 1, 0, 0, 0] P \Rightarrow [0, 0, 3/4, 0, 0, 0, 0]$$

$$p^{(2)} = [0, 0, 3/4, 0, 0, 1/4, 0] P \Rightarrow [0, 9/16, 0, 0, 3/8, 1/16]$$

$$p^{(3)} = [0, 9/16, 0, 0, 3/8, 0, 1/16] P \Rightarrow [27/64, 0, 0, 27/64, 0, 5/32]$$

$$p^{(4)} = [27/64, 0, 0, 27/64, 0, 5/32] P \Rightarrow [27/64, 0, 81/256, 0, 0, 5/256]$$

$$P(\text{absorbed by 4}) = 81/256 + 27/256 + 5/32 = 37/64 = 0.515$$

Tutorial 7; (2)

State Space: $\{M, A, S\}$

$$P = \begin{matrix} & \begin{matrix} M & A & S \end{matrix} \\ \begin{matrix} M \\ A \\ S \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} \end{matrix}$$

In the initial year, 2000, $p^{(0)} = \begin{bmatrix} M & A & S \\ 0 & 0 & 1 \end{bmatrix}$

a) In the year 2002; $p^{(2)} = p^{(1)} p$

$$p^{(1)} = p^{(0)} p \Rightarrow [0 \ 0 \ 1] \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} \Rightarrow [1/3 \ 1/3 \ 1/3]$$

$$p^{(2)} = [1/3 \ 1/3 \ 1/3] \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1/9 & 4/9 & 4/9 \\ M & A & S \end{bmatrix}$$

$\therefore$ The probability that he has Santos in 2002 is $4/9$

b) The probability that he has Maruti in 2002 is $1/9$

c) In the year 2003; $p^{(3)} = p^{(2)} p$

$$\Rightarrow p^{(3)} = [1/9 \ 4/9 \ 4/9] \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} \Rightarrow \begin{bmatrix} 4/27 & 7/27 & 16/27 \\ M & A & S \end{bmatrix}$$

$\therefore$ The probability that he had Ambassador in 2003 is $7/27$

d) The probability that he had Santos in 2003 is $16/27$

ii) To find the probability in the long run, we should find VFPV such that $\sum V = V_1 + V_2 + V_3 = 1$ and $VP = V$

$$\Rightarrow [V_1 \ V_2 \ V_3] \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} = [V_1 \ V_2 \ V_3]$$

$$\Rightarrow [V_3/3 \ V_1 + V_3/3 \ V_2 + V_3/3] = [V_1 \ V_2 \ V_3]$$

$$\Rightarrow V_1 = \frac{V_3}{3}, \quad V_2 = V_1 + \frac{V_3}{3}, \quad V_3 = V_2 + \frac{V_3}{3} \Rightarrow V_1 = \frac{V_3}{3}$$

$$\Rightarrow V = \begin{bmatrix} 1/6 & 1/3 & 1/2 \end{bmatrix} \quad P(S) = 1/2$$

• Absorbing state: In a Markov chain of the process, reaches to a certain state after which it remains in the same state then the state is called absorbing state. In other words, a state ' a_i ' is said to be in absorbing state when $P_{ij} = \begin{cases} 1 & \text{for } i=j \\ 0 & \text{otherwise} \end{cases}$

• Recurrent State: A state ' a_i ' is said to be a recurrent state if starting from state ' a_i ', the system eventually return to the same state.

• Periodic State: A state ' a_i ' is said to be periodic with period ' T ' if return to that state is possible at instants $(T, 2T, 3T, \dots)$

Queueing Theory

• Queue: Queue is a common word which means either a waiting line/an act of joining the line. It is formed when the no. of customers arrival is greater than the no. of customers being served during the period of time.

Example: Patient's waiting in Doctor clinic.

Customers waiting in bank service.

Vehicles waiting in petrol bunk.

In all the above, situations have some common features like customers arrival, formation of queue, waiting for service, prohibition of service to customers in a certain order, Departure after the service etc.

Here, we learn how to construct & solve the equations that describes doing behaviour for the above discussed situations. The main aim of developing queueing

model is to minimize the total cost of queue and the cost of time with the suitable mathematical model.

Basic characteristics of queuing system/model:

1) Arrival pattern

2) Service pattern

3) Queuing discipline

1. Arrival Pattern: The pattern of arrivals like patients, vehicles etc is known as arrival pattern of the queue.

Arrival pattern is either regular/irregular.

Generally it is random and follows poisson's process with mean equal to average no. of arrivals per unit time

Arrival of customers is variance.

2. Service pattern: It is similar to arrival pattern but this is not random because it is interrupted by the idle periods when no customers arrives.

3. Queuing Discipline: The order in which members of the queue are selected for service is known as queuing discipline which can be First In First Out, First Come First Serve, Selection in Random Order.

FIFO or FCFS, LIFO or LCFS, SIRO.

Some Basic Terminologies in Queuing Model:

• Customer → A unit coming to service station.

Ex: persons, books, telephone calls etc.

• Waiting Line → A line formed by customers waiting to receive service.

• Arrival Rate (λ) → No. of customers arrival per unit time. Also called Mean arrival rate.

• Service Rate (μ) → The no. of customers being served per unit time. Also called Mean Service Rate.

n → no. of customers in the system.

m → no. of customers in the queue.

$\frac{1}{\lambda}$ → Mean inter-arrival time.
(Time gap b/w arrivals)

$\frac{1}{\mu}$ → Mean inter-service time.
(Time gap b/w 2 services)

• Symbolic representation of Queuing Model:

Kendall's Notation (a/b/c : d/e)

a → probability distribution of the arrivals of customers per unit time.

b → probability distribution of the service time.

c → no. of servers.

d → capacity of the system (finite / infinite capacity)

e → queue discipline.

1st model > M/M/1 : ∞ / FIFO

In m/m 1 queuing system, the arrival rate follows poisson process i.e., λ and service time is exponential with only 1 server.

This models deals with a queuing system having single server, poisson arrival, exponential service with infinite capacity and the service is on the basis of First In First Out.

• Traffic Intensity (or utilization rate): ' ρ '
(gives probability of server is busy)

$$\rho = \frac{\lambda}{\mu}$$

Traffic Intensity = $\frac{\text{Arrival rate}}{\text{Service rate}}$
 $P(\text{server is busy})$

• Idle rate (P_0): $\{P_0 \rightarrow P(\text{server is free})\}$

$$P_0 = 1 - \rho$$

• Expected number of customers in the system: $[E(n)]$

$$E(n) = \frac{\rho}{1 - \rho} = \frac{\lambda}{\mu - \lambda} = L_s \quad \text{length}$$

• Expected number of customers in the Queue: $[E(m)]$

$$E(m) = E(n) - \rho$$

$$E(m) = \frac{\lambda^2}{\mu(\mu - \lambda)} = L_q$$

* $m \rightarrow$ no. of customers in queue

* $n \rightarrow$ ———— system

* $1/\mu \rightarrow$ mean inter service

* $1/\lambda \rightarrow$ ———— inter-arrival

* $\lambda \rightarrow$ mean arrival rate

* $\mu \rightarrow$ ———— service rate

Expected time spent by the customer in the system: $[E(T)]$

$$E(T) = \frac{E(n)}{\lambda} = \frac{1}{\mu - \lambda} = W_s$$

Average waiting time: $[E(W)]$

$$E(W) = E(T) - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)} = W_q$$

Probability of 'k' members in queue:

$$P(k \text{ members in queue}) = P_m = \rho^k$$

Probability of 'k' members in system:

$$P(k \text{ members in system}) = P_n = \rho^k P_0$$

Probability that no. of customers in system exceeds 'k':

$$P(n > k) = \rho^{k+1}$$

Probability that no. of customers in queue exceeds 'k':

$$P(m > k) = \rho^k$$

Probability of waiting time of customers in system exceeds 't':

$$P(W_s > t) = e^{-(\mu - \lambda)t}$$

Probability of waiting time of customers in queue exceeds 't':

$$P(W_q > t) = \rho e^{-(\mu - \lambda)t}$$

⑥

1. Suppose that customers arrive at a Poisson rate of one per day every 12 minutes and that the service time is exponential at a rate of one service per 8 minutes. What is

a) the average number of customers in the system.

b) the average time a customer spends in the system.

→ Given; $1/\lambda = 12 \text{ minutes} \Rightarrow \lambda = 1/12$

$1/\mu = 8 \text{ minutes} \Rightarrow \mu = 1/8$

a) The average number of customers in the system:

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{1/12}{1/8 - 1/12} = \frac{0.0834}{0.125 - 0.0834} = 2/\text{min}$$

b) The average time a customer spends in the system:

$$E(T) = W_s = \frac{E(n)}{\lambda} \Rightarrow \frac{2}{1/12} = 24 \text{ min}$$

2. A supermarket has a single cashier. During peak hours, customers arrive at a rate of 20 per hour. The average number of customers that can be processed by cashier is 24 per hour. Calculate: a) The probability that the cashier is idle.
b) The average number of customers in the queuing system.
c) The average time a customer spends in the system.
d) The average number of customers in queue.

→ Given; $\lambda = 20$ per hour

$\mu = 24$ per hour

$$\text{W.K.T; } \rho = \frac{\lambda}{\mu} = \underline{\underline{0.84}}$$

e) The average time a customer spends in the queue waiting for service.

a) The probability that the cashier is idle.

$$P_0 = 1 - \rho = \underline{\underline{0.16}}$$

b) The average number of customers in queuing system.

$$E(n) = L_s = \frac{\lambda}{\mu - \lambda} = \frac{20}{24 - 20} = \underline{\underline{5}}$$

c) The average time a customer spends in the system.

$$E(T) = \frac{E(n)}{\lambda} = \frac{5}{20} = \underline{\underline{0.25}} = W_s \text{ (hr)}$$

d) The average number of customers in the queue.

$$E(m) = E(n) - \rho = 5 - 0.16 = \underline{\underline{4.84}} \approx \underline{\underline{4}} \text{ customers}$$

e) The average time a customer spends in the queue waiting for service.

$$E(W) = E(T) - 1/\mu = 0.25 - 0.0416 = \underline{\underline{0.2}} \text{ hr}$$

3. Customers arrive at a sales counter manned by a single person according to a Poisson process with a mean rate of 60 per hour. The time required to serve a customer has an exponential distribution with a mean of 100 seconds. Find the average waiting time of a customer.

→ Given; $\lambda = 60$ per hour

$1/\mu = 100$ seconds. ($60 \times 60 / 100 = 1/100$) $\Rightarrow 36$ per hour

Average waiting time of a customer:

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{36 - 60} = \underline{\underline{0.0625}} \text{ (E(T)) hr}$$

$$W_q = E(T) - 1/\mu = 0.0625 - 0.027 = \underline{\underline{0.0355}} \text{ hr}$$

4. In a city airport, flights arrive at a rate of 24 flights per day. It is known that the inter-arrival time follows an exponential distribution and the service time distribution is also exponential with average of 30 minutes.

→ Given: $\lambda = 24$ per day (1440 min) $\times 24 \Rightarrow 24/24 \times 60 = 1/60$ per min
 $1/\mu = 30$ min $\Rightarrow \mu = 1/30$ per min

a) The probability that the system will be idle:

W.k.t; $\rho = \frac{\lambda}{\mu} = 0.5$

$P_0 = 1 - \rho = 1 - 0.5 = 0.5$

b) The mean queue size: $L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = E(m) = \frac{2.71 \times 10^{-4}}{1/30(0.0166)}$

$E(m) = 0.5$

c) The average number of flights: $L_s = E(n) = \frac{\rho}{1 - \rho} = \frac{0.5}{0.5} = 1$

d) The probability that the size exceeds 7.

$P(n > k) = \rho^{k+1} \Rightarrow P(n > 7) = \rho^8 = (0.5)^8$

$= 3.90 \times 10^{-3}$

$= 0.0039$

5. Customers arrive at a watch repair shop according to a Poisson process at a rate of one per every 10 minutes and the service time is exponential with mean 8 minutes. Find the average number of customers in the shop, average waiting time a customer spends in the shop and the average time a customer spends in the queue for service.

→ Given: $\lambda = 1/10$ per minutes

$\mu = 1/8$ per minutes

W.k.t; $\rho = \frac{\lambda}{\mu} = \frac{1/10}{1/8} = 0.8$

a) Average number of customers in the shop:

$L_s = E(n) = \frac{\rho}{1 - \rho} = \frac{0.8}{1 - 0.8} = 4$

b) Average waiting time a customer spends in shop:

$W_s = \frac{1}{\mu - \lambda} = \frac{1}{1/8 - 1/10} = \left(\frac{2}{3} \times 10\right) = 40 \text{ min}$

c) average waiting time in queue W_q :

$$W_q = \frac{\rho}{\mu - \lambda} = \frac{0.8}{0.025} = 32 \text{ minutes}$$

2nd model $M/M/1: C/FIFO$

This model is the modified form of the previous model. In this model, the maximum no. of customers are limited to 'C'.

P(having n customers in the system):

$$P_n = \begin{cases} \frac{(1-\rho)\rho^n}{1-\rho^{C+1}}, & \rho \neq 1 \\ \frac{1}{C+1}, & \rho = 1 \end{cases}$$

Probability

$\lambda \rightarrow$ arrival rate
 $\mu \rightarrow$ service rate
 $\rho = \frac{\lambda}{\mu}$

Average number of customers in the system:

$$E(n) = L_s = \sum_{n=0}^C n P_n = \begin{cases} \frac{\rho}{1-\rho} - \frac{(C+1)\rho^{C+1}}{1-\rho^{C+1}}, & \rho \neq 1 \\ C/2, & \rho = 1 \end{cases}$$

Idle rate: $P_0 = \frac{1-\rho}{1-\rho^{C+1}}$ for $\rho \neq 1$

Average number of customers in the queue:

$$E(m) = E(n) - \sum_{n=0}^C P_n = L_q \quad \text{or} \quad L_q = L_s - \frac{\lambda'}{\mu}$$

Waiting time in the system: $E(T) = W_s = \frac{E(n)}{\lambda'}$

Waiting time in the queue: $E(W) = \frac{E(m)}{\lambda'}$

where, $\lambda' = \mu(1-P_0)$ is called effective arrival rate.
 $\lambda' = \lambda(1-P_k)$

1. Patients arrive at a doctor's clinic according to Poisson distribution at a rate of 30 patients per hour. The waiting room doesnot accomodate more than 9 patients. Examination time per patient is exponential with mean rate of 20 per hour. Find the following:

→ Given: $C = 9 + 1 = 10$ (9 in waiting room & 1 in service)

$\lambda = 30$ per hour

$\mu = 20$ per hour

W.k.t; $\rho = \frac{\lambda}{\mu} = \frac{3}{2} = 1.5$

a) Probability that an arriving patient will not wait:

$$P_0 = \frac{1 - \rho}{1 - \rho^{C+1}} \Rightarrow \frac{1 - 1.5}{1 - 1.5^{11}} = \frac{-0.5}{-85.5} = 0.005$$

b) Effective arrival rate: $\lambda' = \mu(1 - P_0)$

$$\Rightarrow \lambda' = 20(1 - 0.005) = 19.9 [\approx 20 \text{ per hr}]$$

c) Average number of patients in the clinic:

$$E(n) = \frac{\rho}{1 - \rho} - \frac{(C+1)\rho^{C+1}}{1 - \rho^{C+1}} = \frac{1.5}{-0.5} - \frac{11(1.5)^{11}}{1 - (1.5)^{11}}$$

$$\Rightarrow -3 - \left[\frac{-951.47}{85.5} \right] \Rightarrow -3 + 11.128 = 8.128 \text{ per hr}$$

d) Expected time a patient spends in the clinic:

$$E(T) = W_s = \frac{E(n)}{\lambda'} = \frac{8.128}{19.9} = 0.408 \text{ hr}$$

e) Average number of patients in the queue:

$$E(m) = E(n) - \sum_{n=1}^C P_n \quad \text{or} \quad L_q = 8.128 - \frac{19.9}{20}$$

$$\Rightarrow L_q = 8.128 - 0.995 = 7.133$$

f) Expected waiting time of a patient in the queue:

$$E(W) = \frac{E(m)}{\lambda'} = \frac{7.133}{19.9} = 0.3588 \text{ hr}$$

2. A city has a one person barber shop which can accommodate a maximum of 5 people at a time. (4 waiting & 1 getting haircut). On average, customers arrive at the rate of 8 per hour and the barber takes 6 minutes for serving each customer. It is estimated that the arrival process is Poisson and the service time is an exponential random variable.

→ Given; $C = 5$

$\lambda = 8$ per hour

$\mu = 1/6$ per minute = $1/6 \times 60 = 10$ per hour

W.k.t; $\rho = \frac{\lambda}{\mu} \Rightarrow \rho = \frac{8}{10} = 0.8$

Find.

a) Percentage of time the barber is idle:

$$P_0 = \frac{1-\rho}{1-\rho^{c+1}} \Rightarrow P_0 = \frac{1-0.8}{1-0.8^6} \Rightarrow \frac{0.2}{0.738} = 0.271$$

Percentage: $0.271 \times 100 = 27\%$

b) Effective arrival rate of customers for the shop:

$$\lambda' = \mu(1-P_0) = 10(1-0.271) = 7.29 \text{ per hr}$$

c) Expected number of customers in the barber shop:

$$E(n) = \frac{\rho}{1-\rho} - \frac{(c+1)\rho^{c+1}}{1-\rho^{c+1}} \Rightarrow \frac{0.8}{0.2} - \frac{6(0.8)^6}{1-(0.8)^6}$$

$$\Rightarrow 4 - \frac{1.573}{0.738} \Rightarrow E(n) = 1.868 \text{ per hr}$$

d) Expected time a person spends in the barber shop:

$$E(T) = \frac{E(n)}{\lambda'} = \frac{1.868}{7.29} = 0.2563 \text{ hr}$$

e) Expected number of customers waiting for a hair-cut:

$$\begin{cases} E(W) = \frac{E(m)}{\lambda'} \\ E(m) = L_q = L_s - \frac{\lambda'}{\mu} \end{cases}$$

$$L_q = 1.868 - \frac{7.29}{10} = 1.1394 \text{ per hr}$$

f) Expected waiting time of a customer in the queue:

$$E(W) = \frac{E(m)}{\lambda'} = \frac{1.1394}{7.29} = 0.1563 \text{ hr}$$

3. Trains arrive at the yard every 15 minutes & the service time is 33 minutes. If the line capacity of the yard is limited to 4 trains, then find.

→ Given: $C=4$

$$\lambda = 60/15 = 4 \text{ trains/hour}$$

$$\mu = 60/33 \approx 1.818 \text{ trains/hour}$$

$$\text{Traffic Intensity: } \rho = \lambda/\mu = 4/1.818 \approx 2.2$$

∴ The probability that the yard is empty:

$$P_0 = \frac{1-\rho}{1-\rho^{k+1}} \Rightarrow \frac{1-2.2}{1-(2.2)^5} = \frac{-1.2}{-25.05} \approx 0.0238$$

$$L = \frac{\rho(1 - (k+1)\rho^k + k(\rho^{k+1}))}{(1-\rho)(1-\rho^{k+1})}$$

$$L = \frac{2 \cdot 2(1 - 5(23.4) + 4(51.5))}{(1-2.2)(1-51.5)}$$

$$\Rightarrow L = \frac{2 \cdot 2(1 - 117 + 206)}{(-1.2)(-50.5)} = \frac{2 \cdot 2 \times 90}{60.6} \approx \frac{198}{60.6} = 3.27 \text{ trains.}$$

3rd model > M/M/s: ∞/FIFO

This model is the generalisation of M/M/1: ∞/FIFO, in the sense that service channel is 's' in number instead of single server. Here input is poisson (arrival) & service is exponential. The mean arrival rate is given by $\lambda_n = \lambda \forall n$. If there are more 's' customers in the system, all the servers 's' remain busy and each is of mean rate ' μ '. And hence, the mean service rate is;

$$\mu_n = \begin{cases} n\mu, & 1 \leq n \leq s \\ s\mu, & n > s \end{cases}$$

On the other hand, if there are fewer customers than 's' in the system ($n < s$) only $n(s)$ servers are busy and thus the mean rate is ' $n\mu$ '.

Steady State Solution:

$$P_n = \begin{cases} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n P_0, & 1 \leq n \leq s \\ \frac{1}{s^{n-s} s!} \left(\frac{\lambda}{\mu}\right)^n P_0, & n \geq s \end{cases}$$

Making use of boundary conditions:

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n + \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s \frac{s\mu}{s\mu - \lambda} \right]^{-1}$$

Average Queue Length: $L_q = \frac{\lambda \mu \left(\frac{\lambda}{\mu}\right)^s P_0}{(s-1)!(s\mu - \lambda)^2}$

Average no. of customers in the system: $L_s = L_q + \lambda/\mu$

Average waiting time in the queue: $W_q = \frac{L_q}{\lambda}$

Average waiting time in the system: $W_s = W_q + \frac{1}{\mu}$

1. A travel centre has three service counters to receive people who visit to book air tickets. The customers arrive in a Poisson distribution with the average arrival of 100 persons in a 10-hour service day. It has been estimated that the service time follows an exponential distribution. The average service time is 15 minutes. Find the

→ Given; $\lambda = 100/10 = 10$ per hour

$1/\mu = 15 \Rightarrow \mu = 1/15$ per min $\Rightarrow 60/15 = 4$ per hour

a) Expected number of customers in the system

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s \frac{s\mu}{s\mu - \lambda} \right]^{-1}$$

Here; $s = 3 \Rightarrow$ W.K.T; $\frac{\lambda}{\mu} = \frac{10}{4} = 2.5$

$$\Rightarrow P_0 = \left[\sum_{n=0}^2 \frac{1}{n!} (2.5)^n + \frac{1}{3!} (2.5)^3 \frac{3(4)}{3(4) - 10} \right]^{-1}$$

$$\Rightarrow P_0 = \left[\frac{1(2.5)^0}{0!} + \frac{1(2.5)^1}{1!} + \frac{1(2.5)^2}{2!} + \frac{1}{6} (2.5)^3 \times \frac{12}{12-10} \right]^{-1} = \left[1 + 2.5 + 3.125 + \frac{1}{6} (2.5)^3 \times 2 \right]^{-1}$$

$$\Rightarrow P_0 = (1 + 2.5 + 3.125 + 15.625) (6)^{-1} = (22.25)^{-1} = 0.0449$$

b) $L_q = \frac{\lambda \mu \left(\frac{\lambda}{\mu} \right)^s P_0}{(s-1)! (s\mu - \lambda)^2} = \frac{40 (2.5)^3 (0.0449)}{2(12-10)^2}$

Expected customers in queue, $L_q = 3.507$

a) $L_s = L_q + \frac{\lambda}{\mu} = 3.507 + 2.5 = 6.007$

c) Expected time a customer spends in the system:

$$W_s = W_q + \frac{1}{\mu}$$

$$\Rightarrow W_q = \frac{L_q}{\lambda} = \frac{3.507}{10} = 0.3507$$

$$\Rightarrow W_s = 0.3507 + 0.25 = 0.607$$

d) Expected waiting time for a customer in queue:

$$W_q = 0.3507$$

e) Probability that a customer must wait before he gets service

$$P_n = \begin{cases} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n P_0, & 1 \leq n \leq S \\ \frac{1}{S!} \left(\frac{\lambda}{\mu}\right)^S \left(\frac{S\mu}{S\mu - \lambda}\right) P_0, & n \geq S \end{cases}$$

$$\Rightarrow P(n \geq 3) = \frac{1}{8} \left[2.5 \right]^3 \left[\frac{1.2}{2} \right] (0.449) = \underline{\underline{0.70156}}$$

2. A local hospital has 3 doctors for treating patients. The patients arrive at the hospital according to Poisson distribution with an average of 8 per hour. On an average, a doctor takes about 15 minutes to treat each patient and the actual time taken is known to vary approximately exponentially around this average. Find.

+ Given; $\lambda = 8$ per hour. $S = 3$

$$\frac{1}{\mu} = 15 \Rightarrow \mu = \frac{1}{15} \text{ per min} \Rightarrow \frac{60}{15} = 4 \text{ per hour.}$$

a) Traffic intensity of the system:

$$\rho = \frac{\lambda}{S\mu} = \frac{8}{3(4)} = \frac{2}{3}$$

$$\frac{\lambda}{\mu} = \frac{8}{4} = 2$$

b) Probability that a customer has to wait for service:

$$P_n = \frac{1}{S!} \left(\frac{\lambda}{\mu}\right)^S \left(\frac{S\mu}{S\mu - \lambda}\right) P_0, \quad n \geq S$$

$$P_0 = \left[\sum_{n=0}^{S-1} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n + \frac{1}{S!} \left(\frac{\lambda}{\mu}\right)^S \frac{S\mu}{S\mu - \lambda} \right]^{-1}$$

$$P_0 = \left[\frac{1}{0!} (2)^0 + \frac{1}{1!} (2)^1 + \frac{1}{2!} (2)^2 + \frac{1}{3!} (2)^3 \times \frac{1}{1 - 2/3} \right]^{-1}$$

$$P_0 = \left[1 + 2 + \frac{8}{2} + \frac{8}{3} \right]^{-1} = \frac{1}{9} = \underline{\underline{0.111}}$$

$$\therefore P_n = \frac{1}{S!} (2)^3 \left(\frac{1.2}{4} \right) (0.111) = \underline{\underline{0.444}} = P(n \geq 3)$$

c) Average number of customers waiting in Queue:

$$L_q = \frac{\lambda \mu \left(\frac{\lambda}{\mu}\right)^S P_0}{(S-1)! (S\mu - \lambda)^2} \Rightarrow \frac{3 \cdot 2 (2)^3 (0.111)}{2(4)^2} = \frac{10.656}{32}$$

$$L_q = \underline{\underline{0.89}}$$

d) Average number of customers in the system;

$$L_s = L_q + \frac{\lambda}{\mu} = \underline{2.89} \left\{ 0.89 + \frac{2}{4} \right\}$$

e) Average waiting time a customer spends in the queue

$$W_q = \frac{L_q}{\lambda} = \frac{0.89}{2} = \underline{0.445}$$

f) Expected time a customer spends in the system:

$$W_s = W_q + 1/\mu = 0.445 + \frac{1}{4} = \underline{0.695}$$

3. A bank has three teller counters. The customers to the bank choose a teller counter at random. It is given that the arrivals to the teller counters are Poisson-distributed at the average rate ' λ ' and the mean service rate at the teller counters is $\lambda/2$. Find the steady average queue at each counter.

→ Given; $S = 3$, $\mu = \frac{\lambda}{2} \Rightarrow \frac{\lambda}{\mu} = 2$

Steady average queue;

$$P_0 = \left[\sum_{n=0}^{S-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{S!} \left(\frac{\lambda}{\mu} \right)^S \frac{S!}{S! - \lambda} \right]^{-1}$$

$$\text{w.k.t; } \rho = \frac{\lambda}{\mu} \Rightarrow \frac{\lambda/3}{\lambda/2} = \underline{2/3}$$

$$L_q = \frac{\rho^2}{1-\rho} \Rightarrow \frac{(2/3)^2}{1-2/3} \Rightarrow \frac{4/9}{1/3} = \frac{4}{3} \approx 1.33 \text{ is the average queue length.}$$